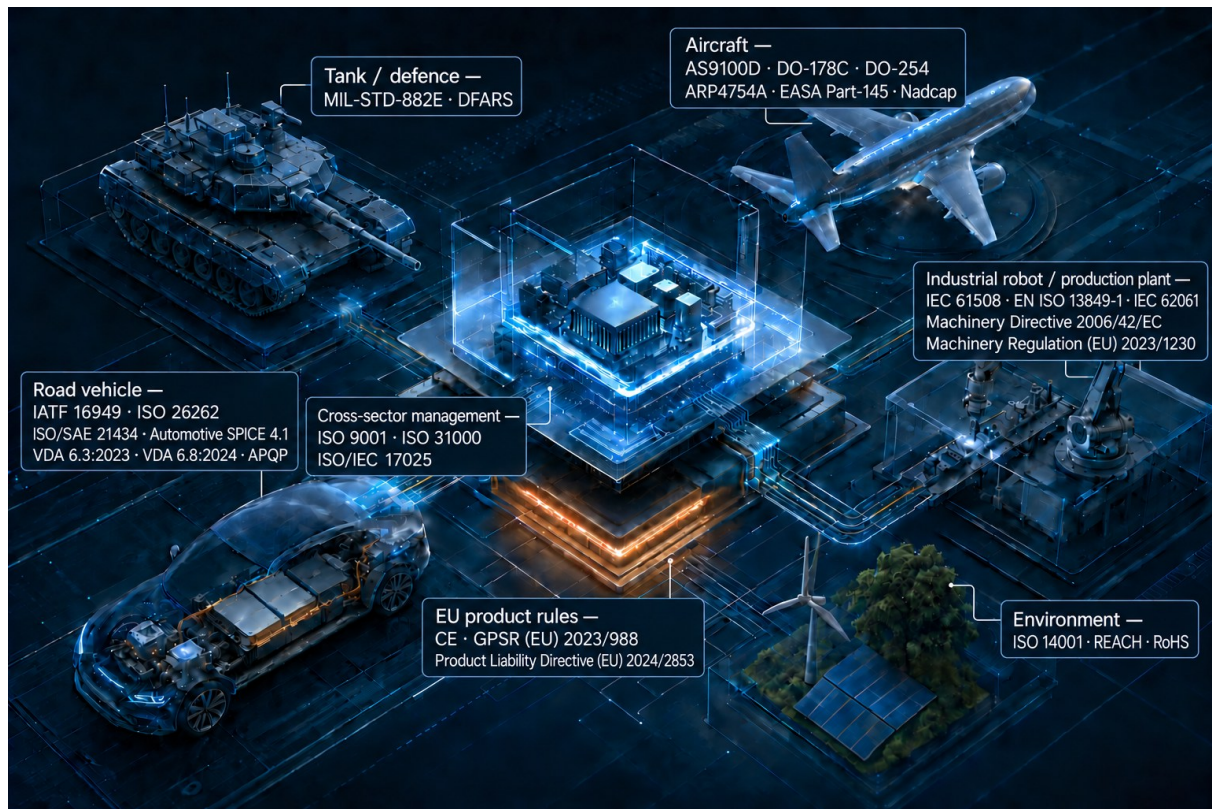


Standards, Regulatory Requirements, and Quality Management in Highly Regulated Industries



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Automotive | Aerospace | Industry | Defense

1 Introduction

Companies in highly regulated industries must do more than manufacture technically capable products. They must also demonstrate that development, manufacturing, the supply chain, documentation, and approval processes meet the applicable legal, standards-based, and contractual requirements.

This analysis, current as of September 6, 2026, highlights the need for an integrated quality and compliance management system that consistently controls shared processes while ensuring that industry- and product-specific evidence is available.

2 Analysis of Standards and Regulatory Requirements

The standards and regulations covered address quality management, laboratory and testing competence, functional safety, aircraft certification and maintenance, critical special processes, U.S. defense contracts, environmental management, and European product law. They differ in legal effect, scope, and evidence requirements. Noncompliance may lead to shipment holds, recalls, certification delays, loss of contracts, regulatory action, and liability risks.

Scope and Disclaimer: This whitepaper provides professional orientation. It does not replace a review of the specific scope of application and binding original texts, or individual legal or certification advice

2.1 Automotive and the Automotive Supply Chain

2.1.1 IATF 16949:2016



Core Focus: IATF 16949 incorporates the requirements of ISO 9001:2015 and adds automotive-specific quality management requirements. Key priorities include defect prevention, robust development and manufacturing processes, continual improvement, and compliance with customer-specific requirements.¹

Significance: The standard is relevant to the development and series production of control and power electronics, electric drive systems, and energy storage solutions. Key methods and records include APQP, FMEA, Control Plans, PPAP, MSA, SPC, and 8D reports.

Specific Risk in Case of Noncompliance: Insufficient monitoring of critical products or process characteristics can result in the shipment of defective products. Possible consequences include shipment holds, recalls, recourse claims, and loss of certification. IATF certification is frequently a customer- or OEM-specific requirement for admission to and continued inclusion in the supplier base; the specific obligation must be assessed case by case.

2.1.2 ISO 26262



Core Focus: ISO 26262 addresses the functional safety of electrical and electronic systems in road vehicles. It covers the entire safety lifecycle, from hazard analysis through development, production, and decommissioning.

¹ Adapted from <https://www.mobile2b.com/audits-inspections/iatf-16949>

Significance: Safety goals, ASIL classifications, safety requirements, and verification evidence must be established for safety-related control units, inverters, drive systems, and energy storage systems.

Specific Risk in Case of Noncompliance: An incorrect safety assessment may result in missing safety mechanisms. Consequences can include accidents, personal injury, recalls, product liability claims, and loss of customer approval.

2.1.3 ISO/SAE 21434:2021



Core Focus: Cybersecurity engineering for electrical and electronic systems in road vehicles throughout the product lifecycle.

The emphasis is on protecting vehicle functions and data against unauthorized interference.

Significance: The standard requires systematic identification, assessment, and treatment of cybersecurity risks. It links organizational provisions with requirements for development, production, operation, maintenance, and decommissioning and complements functional safety under ISO 26262.

Specific Risk in Case of Noncompliance: Undetected vulnerabilities may enable manipulation of vehicle functions, unauthorized access to vehicle data, or failure of safety-related systems. Other consequences include rejection by vehicle manufacturers, insufficient evidence in type-approval and customer processes, costly redevelopment, and recalls.

2.1.4 Automotive SPICE



Core Focus: Assessment and improvement of development processes for software-based and electronic vehicle systems.

Significance: Automotive SPICE defines a process reference model and a process assessment model. Process capability is assessed against defined capability levels. Key areas include requirements analysis, architecture, implementation, verification, project management, configuration and change management, and traceability. The model version and assessment scope agreed for the project are decisive.

Specific Risk in Case of Noncompliance: Poorly controlled processes can cause conflicting requirements, missing traceability, undetected development defects, and inadequate verification. Further consequences include poor assessment results, customer escalations, delayed approvals, costly rework, and loss of development contracts.

2.1.5 VDA 6.3:2023 - Process Audits



Core Focus: Assessment of the quality capability of product development and production processes in the automotive supply chain.

Significance: VDA 6.3 provides a structured method for potential analysis and process audits. It covers project management, product and process development, supplier management, series production, and customer support. The aim is to identify process risks early and assess the effectiveness of defined actions.

Specific Risk in Case of Noncompliance: Uncontrolled processes can cause quality defects, unstable production, delivery failures, and defective products. A poor audit rating may lead to repeat audits, customer escalations, special measures, supplier blocks, exclusion from new business, and rework, warranty, and recall costs.

2.1.6 VDA 6.8:2024 - Supply Chain Process Audit



Core Focus: Assessment, assurance, and continual improvement of logistics processes throughout the automotive supply chain.

Significance: VDA 6.8 provides a uniform audit method for project management, supplier and service provider management, procurement logistics, transportation, in-house logistics, and customer management. It supports systematic identification of supply risks, interface problems, and weaknesses in material and information flows.

Specific Risk in Case of Noncompliance: Poorly managed supply chains can cause part shortages, incorrect deliveries, inventory discrepancies, traceability gaps, and production stoppages. Further consequences include expedited shipments, high inventory and logistics costs, customer escalations, contractual penalties where agreed, poor supplier ratings, and loss of follow-on business.

2.1.7 APQP – Advanced Product Quality Planning



Core Focus: Advance planning and assurance of product and process quality from the concept phase through stable series production.

Significance: APQP coordinates quality planning across functions and connects customer requirements, scheduling, product and process development, risk analyses, validation, and approval activities. The third AIAG edition adds sourcing, change management, program met-

rics, risk mitigation plans, and decision gates. The Control Plan is addressed in a separate manual and remains part of coordinated quality planning.

Specific Risk in Case of Noncompliance: Inadequate advance quality planning may result in requirements, risks, or manufacturability issues being identified only shortly before production starts. Possible consequences include denied PPAP or PPF approval, schedule delays, launch problems, scrap, rework, field failures, warranty costs, recalls, and loss of customer confidence.

2.2 Aviation and Space

2.2.1 AS9100D / IAQG 9100:2016



Core Focus: AS9100 includes all ISO 9001:2015 requirements and adds requirements for quality management systems in aviation, space, and defense. Particular priorities include product safety, configuration management, risk management, traceability, and counterfeit part prevention.

Significance: For electronic aerospace components, development baselines, bills of materials, serial numbers, tests, approvals, and changes must be clearly documented and traceable.

Specific Risk in Case of Noncompliance: If the configuration of a delivered aerospace product cannot be clearly demonstrated, the customer may block or reject the shipment. Certification findings and loss of aerospace contracts are additional risks.

2.2.2 ARP4754A and ARP4754B



Core Focus: Systematic development of civil aircraft and complex airborne systems. ARP4754A addresses development assurance, function and requirements allocation, system architecture, integration, validation, verification, and coordination with safety analyses.

Significance: The guidance links aircraft-level safety objectives to the development of systems, hardware, and software. It provides an important connection to DO-178C and DO-254 and determines the required development and evidence effort based on criticality.

Specific Risk in Case of Noncompliance: Incorrect function allocation, incomplete requirements, incorrect Functional or Item Development Assurance Level (FDAL/IDAL) assignments, or missing evidence may leave safety-critical system interactions undetected. When ARP4754A is agreed with the certification authority as a means of compliance or is contractually required, consequences may include rejection of evidence, redesign, repeated testing, certification delays, additional costs, and loss of contracts.

Edition Note: The FAA recognizes ARP4754A through active AC 20-174 as an acceptable method for development assurance. SAE published its successor, ARP4754B, in December 2023. The applicable edition must be coordinated for the specific project and certification context.

2.2.3 DO-178C



Core Focus: DO-178C describes requirements for the development and verification of airborne software. Complete traceability among requirements, software design, source code, and tests, together with controlled configuration and quality management, is essential.

Significance: Agile software development must be integrated into the development process so that all safety-related requirements, changes, tests, and approvals are documented and traceable.

Specific Risk in Case of Noncompliance: Missing traceability or test evidence can result in denied or delayed software approval. This may cause extensive rework, project delays, and additional costs.

2.2.4 DO-254



Core Focus: DO-254 addresses the development of complex electronic hardware for airborne systems, particularly FPGAs, Application-Specific Integrated Circuits (ASICs), Programmable Logic Devices (PLDs), and comparable programmable hardware.

Significance: When developing proprietary aerospace electronics, hardware requirements, architecture, implementation, verification, configuration, and approval must be fully documented.

Specific Risk in Case of Noncompliance: Insufficient verification of a safety-critical hardware function can lead to rejection of certification evidence. This can result in redesign, repeated testing, and significant schedule delays.

2.2.5 EASA Part-145



Core Focus: Approval and oversight of organizations that maintain aircraft and aeronautical components.

Significance: EASA Part-145 establishes requirements for organization, personnel, facilities, tools, maintenance data, release procedures, safety management, and documentation. Its purpose is to ensure safe and compliant performance and certification of maintenance.

Specific Risk in Case of Noncompliance: Incorrect or incompletely documented maintenance can compromise airworthiness. Possible consequences include findings, restricted or

revoked approvals, denied releases to service, operational interruptions, flight cancellations, and civil, regulatory, or criminal consequences.

2.2.6 Nadcap



Core Focus: Industry-recognized accreditation of critical special processes in aviation, space, and defense.

Significance: Nadcap assesses areas including heat treatment, chemical processing, nondestructive testing, welding, coatings, materials testing, and electronics manufacturing. Audits verify compliance with technical industry requirements and customer-specific provisions.

Specific Risk in Case of Noncompliance: Nonconforming special processes may cause undetected material or component defects. Further consequences include denial, suspension, or withdrawal of accreditation, removal from approved supplier lists, lost orders, repeat audits, rework, and production interruptions.

2.3 Military and Defense

2.3.1 MIL-STD-882E



Core Focus: Systematic identification, assessment, and reduction of hazards in military systems throughout their lifecycle.

Significance: When contractually or programmatically required, the standard applies to safety-related electronic and programmable systems in U.S. defense projects, including safety plans, hazard analyses, safety requirements, verification, and formal residual risk acceptance.

Specific Risk in Case of Noncompliance: Inadequately controlled hazards may lead to denied system approval, contract breaches, redesign, delivery delays, personal injury, and property damage.

2.3.2 DFARS



Core Focus: DFARS contains contractual and legal requirements for suppliers to the U.S. Department of Defense. These include cybersecurity, protection of controlled information, material origin, specialty metals, supplier evidence, and counterfeit part prevention.

Significance: For defense contracts, the applicable contractual DFARS clauses must be reviewed and implemented in purchasing, IT, manufacturing, documentation, and supplier management.

Specific Risk in Case of Noncompliance: Missing origin records or prohibited materials can lead to rejected shipments, payment holds, contractual penalties, or loss of the defense contract.

2.4 Industry and Machinery Safety

2.4.1 IEC 61508



Core Focus: IEC 61508 is a cross-industry basic standard for the functional safety of electrical, electronic, and programmable electronic systems. It governs the safety lifecycle and the assignment of Safety Integrity Levels.

Significance: The standard is particularly relevant to industrial controls, safety functions, and electronic platforms used across industries.

Specific Risk in Case of Noncompliance: If an insufficient SIL is assigned to a safety function or its failure probability is not demonstrated, the function may fail in a hazardous situation. Possible consequences include personal injury, plant shutdowns, and liability claims.

2.4.2 Machinery Directive 2006/42/EC



Core Focus: The Machinery Directive governs the safety of machinery, safety components, and partly completed machinery when first placed on the market or put into service in the European Economic Area. Key elements include risk assessment, essential health and safety requirements, and evidence of product conformity.

Significance: The directive is relevant when a company develops complete machinery, safety components, or partly completed machinery and places these products on the market. For machinery controls, the specific scope of application must be assessed based on their function and product type.

Specific Risk in Case of Noncompliance: For example, placing machinery controls on the market without adequate risk assessment or with an inadequately designed safety function can cause malfunctions and personal injury. Possible consequences include sales bans, authority-ordered recalls, correction or removal of improper CE marking, damages claims, and criminal consequences.

2.4.3 Machinery Regulation (EU) 2023/1230



Core Focus: The Machinery Regulation governs the safety of machinery and related products and their uniform conformity assessment in the European single market. It focuses on risk assessment, essential health and safety requirements, and evidence of product conformity. It replaces Machinery Directive 2006/42/EC from January 20, 2027.

Significance: The regulation applies to newly marketed machinery from January 20, 2027. It addresses, among other matters, safety-related functions using artificial intelligence, protection of safety-related controls and software data against manipulation, and specific procedures for certain higher-risk categories of machinery.

Specific Risk in Case of Noncompliance: Incomplete risk assessment or incorrect conformity assessment may prevent machinery from being placed on the market or put into service. Further consequences may include authority-ordered shutdowns, sales bans, recalls, retrofits, fines, and civil or, where applicable, criminal consequences following personal injury or property damage.

2.5 Environmental Management and Chemicals Legislation

2.5.1 ISO 14001:2026



Core Focus: ISO 14001 governs the establishment of an environmental management system. Companies must systematically manage relevant environmental aspects, legal obligations, environmental objectives, risks, and emergency measures.

Significance: Particular considerations include energy consumption, hazardous substances, electronic waste, emissions, packaging, disposal, and environmental requirements for suppliers.

Specific Risk in Case of Noncompliance: Violations of environmental law, such as improper handling of hazardous substances or waste, may cause environmental damage, regulatory action, fines, and remediation costs. Nonconformity with ISO 14001 can lead to certification findings or loss of certification; the standard itself is not a law.

2.5.2 REACH (EC) No. 1907/2006



Core Focus: Registration, evaluation, authorization, and restriction of chemicals to protect human health and the environment.

Significance: REACH requires companies to collect information on substances used or placed on the market and to control their risks. For product manufacturers, substance restrictions, authorization requirements, substances of very high concern, and supply-chain information duties are particularly relevant.

Specific Risk in Case of Noncompliance: Unregistered, prohibited, or restricted substances can lead to marketing or use bans. Other possible consequences include shipment holds, product withdrawals, regulatory orders, fines, liability claims, and costly material changes during series production.

2.5.3 RoHS Directive 2011/65/EU



Core Focus: Restriction of hazardous substances in electrical and electronic equipment.

Significance: RoHS restricts the use of ten hazardous substances or substance groups, including lead, cadmium, mercury, hexavalent chromium, certain brominated flame retardants, and phthalates. Compliance must be demonstrated through appropriate supplier evidence, material declarations, testing, and technical documentation.

Specific Risk in Case of Noncompliance: If permitted limits are exceeded, the affected product generally cannot lawfully be made available on the European market. Risks include sales bans, withdrawals, recalls, fines, replacement of nonconforming components, and correction of the EU declaration of conformity and CE marking.

2.6 EU Product Law and Product Liability

2.6.1 GPSR (EU) 2023/988



Core Focus: Safety of consumer products, including products offered online and through distance selling.

Significance: The GPSR requires, among other things, an internal risk analysis, appropriate technical documentation, clear product and manufacturer identification, understandable safety information, and effective processes for complaints, safety incidents, and recalls. It applies particularly where specific European product legislation does not fully cover a risk.

Specific Risk in Case of Noncompliance: Market surveillance authorities may challenge or remove an unsafe or inadequately documented product from the market. Possible consequences include sales bans, withdrawals, public recalls, Safety Gate notifications, fines, damages claims, and significant reputational harm.

2.6.2 Other EU Product Requirements



Core Focus: In addition to product-specific harmonization legislation, requirements concerning market surveillance, general product safety, cybersecurity, chemicals, and product liability must be assessed as applicable to the product.

Significance: Duties extend from design and manufacturing through placing on the market and post-market monitoring to corrective actions, withdrawals, recalls, and disposal. Directive (EU) 2024/2853 modernizes product liability, particularly for software, digital functions, and updates.

Specific Risk in Case of Noncompliance: Unclear responsibilities and incomplete evidence increase the risk of regulatory action, recalls, liability cases, and reputational damage.

2.7 Cross-Industry Management and Testing Competence

ISO 9001 provides the basis for shared quality management processes; ISO 31000 provides cross-industry risk management guidelines. Their integrated application is explained in the following sections. ISO/IEC 17025 complements these foundations with demonstrated competence of testing and calibration laboratories.

2.7.1 ISO/IEC 17025:2017



Core Focus: Competence, impartiality, and consistent operation of testing and calibration laboratories.

Significance: The standard establishes requirements for personnel, testing and calibration methods, equipment, metrological traceability, validation of results, and reporting. It builds confidence in the technical validity and comparability of laboratory results.

Specific Risk in Case of Noncompliance: Incorrect or nontraceable results can lead to incorrect product releases and safety decisions. Further risks include rejection of test reports, repeat testing, loss of customers, and restriction, suspension, or withdrawal of accreditation.

3 Integrated Implementation of Standards and Regulations

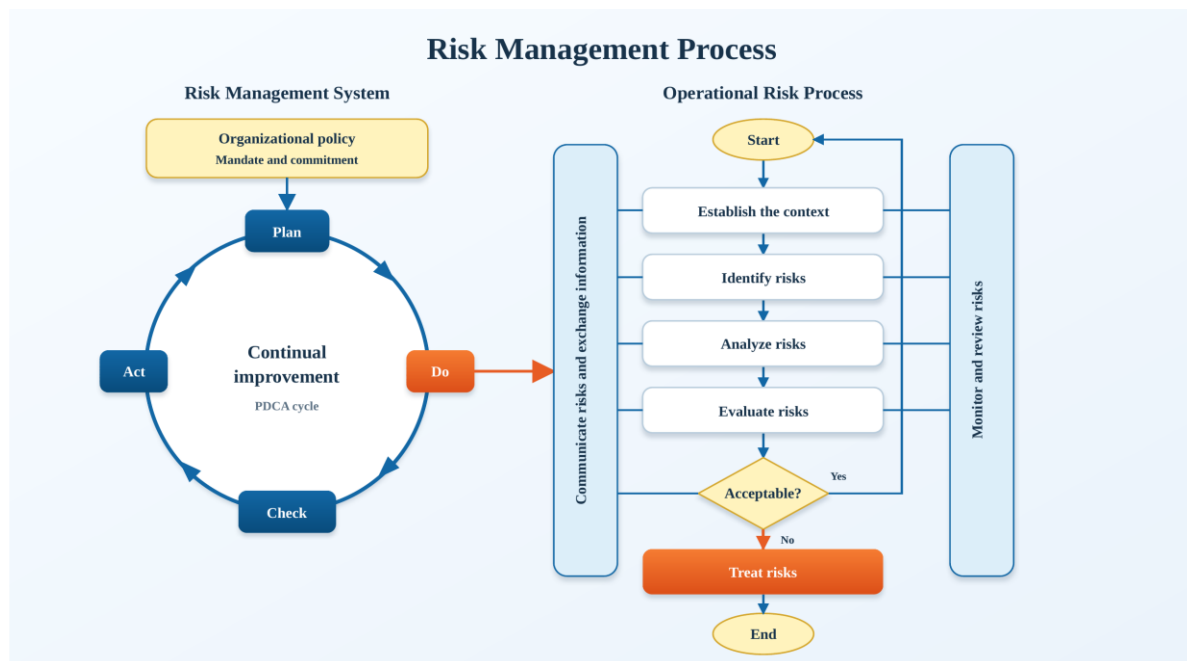
These standards and regulations should not be implemented in isolated management systems. Companies should use shared core processes for document control, risk and change management, supplier management, competence management, internal audits, and management review. Responsibilities, acceptance criteria, and evidence remain specific to each standard or regulation. Together, these elements form an integrated system.

For automotive projects, Automotive SPICE, APQP, VDA 6.3, and VDA 6.8 can be connected through shared requirements, change, supplier, and approval processes. In aerospace, ARP4754A provides the link to systems development. A mapping matrix should identify the relevant standard or regulation, applicable edition, responsible role, and required evidence for each requirement.

4 The Risk-Based Approach

A risk-based approach enables early detection and assessment of deviations and their prevention through appropriate controls. Risks are therefore considered during planning, development, procurement, manufacturing, delivery, and change, rather than being addressed only after a failure occurs.

The basis is DIN EN ISO 9001:2015-11, Section 6.1, "Actions to address risks and opportunities." ISO 9001 explicitly connects the process approach with the Plan-Do-Check-Act (PDCA) cycle and risk-based thinking. DIN ISO 31000:2018-10, "Risk management - Guidelines," provides complementary guidance.



The Risk Management Process in a Risk-Based Approach²

The applicable assessment criteria depend on the specific standard or regulation. Quality risks often focus on severity, occurrence, and detection. Functional safety adds criteria such as exposure, controllability, safety integrity, and permissible residual risk. For legal and contractual obligations, it is also necessary to determine whether any deviation is permissible at all.

4.1 Quality and Process Requirements

IATF 16949 and AS9100 use risks primarily to prioritize quality, process, and supplier actions. Not every process is monitored with the same intensity. Processes with a major impact on product safety, delivery capability, or customer satisfaction receive stricter controls.

Under IATF 16949, product, process, supplier, and customer risks are proactively assessed and controlled through APQP, FMEA, Control Plans, SPC, MSA, and contingency planning. The 8D method provides a structured response to problems that have already occurred. Limited residual quality risk may be accepted if customer, product, and safety requirements remain satisfied and the decision is documented. Critical characteristics and processes receive stricter controls.

²Source of the basic structure: Brühwiler, Bruno (2007): Risikomanagement als Führungsaufgabe [Risk Management as a Management Responsibility]. Haupt Verlag, Bern/Stuttgart/Vienna. The illustration was subsequently supplemented with elements of ISO 31000, first published in 2009, and DIN ISO 31000:2018-10.

AS9100 focuses particularly on product safety, configuration, suppliers, counterfeit parts, and delivery and quality performance. Changes require more intensive review and approval. Residual risks must be documented, monitored, and coordinated with customers and aviation authorities when safety-related effects are involved.

Automotive SPICE directs process improvement toward demonstrated process capability. VDA 6.3 assesses risks in product development and production, while VDA 6.8 addresses logistics. APQP connects early risk analysis with coordinated actions and approval milestones. Assessments, process audits, and advance quality planning complement one another but do not replace product-specific safety evidence.

ISO/IEC 17025 applies the risk-based approach to the validity of test and calibration results. Nadcap focuses it on the control of special critical processes. In both cases, competent personnel, suitable methods, traceable records, and effective corrective actions are essential.

4.2 Safety Standards and Development Assurance

These determine development and evidence effort according to the potential safety impact of a failure. Higher criticality requires stricter development, verification, independence, documentation, and approval. Residual risk may be accepted only after all required safety measures and evidence have been completed.

ARP4754A links aircraft- and system-level safety objectives with requirements allocation, architecture, and development assurance. Software evidence under DO-178C and hardware evidence under DO-254 are coordinated accordingly; the standards are not interchangeable.

The necessary development and verification effort is determined by functional safety requirements. As criticality increases, so do the requirements for traceability, testing, reviews, and independent verification, validation, or confirmation measures.

Standard or Regulation	Safety or Classification Level	Levels from Low to High
DO-178C / DO-254	Design Assurance Level (DAL)	DAL E → DAL D → DAL C → DAL B → DAL A
ISO 26262	Automotive Safety Integrity Level (ASIL)	QM → ASIL A → ASIL B → ASIL C → ASIL D
IEC 61508	Safety Integrity Level (SIL)	SIL 1 → SIL 2 → SIL 3 → SIL 4
MIL-STD-882E	Risk level derived from severity and probability of occurrence	Low → Medium → Serious → High
Machinery Directive 2006/42/EC / Machinery Regulation (EU) 2023/1230	No separate safety levels; safety-related controls are classified under EN ISO 13849-1 or IEC 62061	PL a → PL b → PL c → PL d → PL e, or SIL 1 → SIL 2 → SIL 3
ARP4754A	Functional Development Assurance Level (FDAL) and Item Development Assurance Level (IDAL)	E → D → C → B → A

Table: Classification Levels in Safety Standards

Notes

- DAL, ASIL, SIL, PL, and military risk categories cannot be directly equated.
- In ISO 26262, QM identifies areas that are not assigned an ASIL and are addressed through quality management requirements.

Consequences of disregarding safety standards arise because they may serve as contractual requirements, a basis for certification, recognized state of the art, or evidence of proper development. If a defective product causes personal injury or property damage, the manufacturer may be liable under the German Product Liability Act.

If identifiable safety risks are negligently ignored and people are injured or killed as a result, responsible individuals may be investigated for negligent bodily injury or negligent homicide.

4.3 Legally or Contractually Binding Compliance Requirements

DFARS, EASA Part-145, environmental legal obligations addressed through environmental management, the Machinery Directive, and EU product legislation do not permit arbitrary risk trade-offs. A low probability of occurrence or high implementation costs do not justify ignoring a binding requirement. Nadcap requirements are also binding where agreed contractually or through customer requirements. Risk assessment determines the required controls, not whether the obligation must be met.

4.4 The Difference Between Quality, Safety, and Compliance Risks

Risk assessment answers three different questions:

In a quality standard:

- Which risks matter, and how intensively must they be addressed?
- Quality risks are prioritized according to their significance.

In A safety standard:

- Which technical measures are necessary to limit residual risk to a tolerable level?
- Define formal risk control and establish the necessary safety and evidence effort.
- Safety risks must be reduced to a demonstrably acceptable residual level.

In a legal or contractual compliance requirement:

- Which controls are needed to reliably prevent violations of binding requirements?
- Legal and contractual obligations must be met regardless of probability of occurrence, cost, or economic benefit.

4.5 Example for an Electronics Company

The company is developing an inverter intended for use in a commercial vehicle, an industrial machine, and a military application.

- Under IATF 16949, the risk of defective solder joints or variable manufacturing processes is addressed through FMEA, SPC, MSA, and a Control Plan.
- Under ISO 26262, the assessment determines whether a malfunction could cause a hazard. The required ASIL is derived from severity, exposure, and controllability.
- Under IEC 61508, the required SIL for the industrial safety function is determined and the necessary risk reduction is demonstrated.
- For military use subject to MIL-STD-882E, a hazard analysis is performed, and the remaining risk is accepted by the competent authority.
- Under DFARS, compliance with contractual requirements concerning material origin, cybersecurity, Controlled Unclassified Information (CUI), and subcontractors is also verified.
- When the product is placed on the European market, the applicable EU product requirements must also be met, including the Machinery Directive through January 19, 2027, where applicable; the Machinery Regulation applies from January 20, 2027.

4.6 Conclusion

An integrated quality, safety, and compliance management system is appropriate for companies operating in multiple highly regulated sectors. Shared processes such as risk management, document control, configuration management, change management, supplier management, verification, and auditing should be organized consistently. Industry-specific evidence must remain distinct and clearly traceable.

5 Sources and Publication Status

The following selection identifies the official information sources relevant to this publication. Complete standards must be obtained in their applicable editions from the publishers responsible.

Standard or Regulation	Status as of September 6, 2026	Official Source
AIAG APQP, 3rd Edition	Advance quality planning; 2024. Control Plan published as a separate manual	AIAG
ARP4754A / FAA AC 20-174	2010 Edition A; recognized by active FAA AC 20-174. Check project-specific applicability	FAA
ARP4754B	SAE successor edition, December 2023; does not automatically replace the agreed Edition A	SAE
Automotive SPICE 4.1	Process reference and assessment model; August 2026	VDA QMC
DFARS	Continuously updated contractual requirements	Acquisition.gov
DO-178C / DO-254	Published certification guidance	RTCA
EASA Part-145	Current requirements under Regulation (EU) No. 1321/2014 and EASA Easy Access Rules	EASA
EN ISO 13849-1:2023	Current published edition	ISO
GPSR (EU) 2023/988	Applicable since December 13, 2024	EUR-Lex
IAQG 9100:2016 / AS9100D	Published edition that remains applicable	IAQG
IATF 16949:2016	Published edition that remains applicable	IATF
IEC 61508:2010 (Series)	Second edition; remains published	IEC
IEC 62061:2021 AMD1:2024 + AMD2:2026	⁺ Current consolidated version	IEC
ISO 14001:2026	Published since April 2026; 2015 edition withdrawn	ISO
ISO 26262:2018 (Series)	Current published edition	ISO
ISO 31000:2018	Current published edition	ISO
ISO 9001:2015/Amd 1:2024	Current; successor edition of ISO 9001 in the publication process	ISO
ISO/IEC 17025:2017	Current published edition	ISO
ISO/SAE 21434:2021	Current published edition	ISO

Standard or Regulation	Status as of September 6, 2026	Official Source
Machinery Directive 2006/42/EC	Applicable through January 19, 2027	EUR-Lex
Machinery Regulation (EU) 2023/1230	Applicable from January 20, 2027	EUR-Lex
MIL-STD-882E, Change 1	Current change level, 2023	DLA Quick Search
Nadcap	Continuously updated accreditation program for critical special processes	PRI
Product Liability Directive (EU) 2024/2853	National transposition by December 9, 2026	EUR-Lex
REACH (EC) No. 1907/2006	Consolidated version of June 22, 2026	EUR-Lex
RoHS Directive 2011/65/EU	Applicable as amended	EUR-Lex
VDA 6.3:2023	Process audit; 4th edition, January 2023	VDA QMC
VDA 6.8:2024	Supply chain process audit; December 2024	VDA

6 List of Abbreviations

Abbreviation	Meaning
8D	Eight Disciplines; structured problem-solving and complaint-handling method
AC	Advisory Circular; FAA guidance describing acceptable means of compliance
AIAG	Automotive Industry Action Group; publisher of the APQP manual and other guidance
AMD	Amendment; modification or addition to a standard
APQP	Advanced Product Quality Planning
ARP	Aerospace Recommended Practice; technical guidance published by SAE International
ARP4754A	Guidelines for Development of Civil Aircraft and Systems, Edition A
AS9100D	U.S. edition of IAQG 9100:2016 for quality management in aviation, space, and defense
ASAM	Association for Standardization of Automation and Measuring Systems
ASIC	Application-Specific Integrated Circuit
ASIL	Automotive Safety Integrity Level; safety integrity classification under ISO 26262
ASPICE	Automotive SPICE; process reference and assessment model for automotive development
AUTOSAR	AUTomotive Open System ARchitecture; development partnership for standardized automotive software architectures
CE	CE marking; indicates conformity with applicable EU harmonization legislation
CUI	Controlled Unclassified Information
DAL	Design Assurance Level; development and evidence level under DO-178C and DO-254
DFARS	Defense Federal Acquisition Regulation Supplement
DIN	Deutsches Institut für Normung e. V. (German Institute for Standardization); also the designation of German standards
DLA	Defense Logistics Agency
DO-178C	Software Considerations in Airborne Systems and Equipment Certification
DO-254	Design Assurance Guidance for Airborne Electronic Hardware
E/E	Electrical/electronic
EASA	European Union Aviation Safety Agency

Abbreviation	Meaning
EC	European Community; designation used in older EU legal acts
EN	European Standard
EU	European Union
FAA	Federal Aviation Administration; U.S. aviation authority
FDAL	Functional Development Assurance Level; development assurance classification at the function level
FMEA	Failure Mode and Effects Analysis
FPGA	Field-Programmable Gate Array
GPSR	General Product Safety Regulation; Regulation (EU) 2023/988
IAQG	International Aerospace Quality Group
IATF	International Automotive Task Force
IDAL	Item Development Assurance Level; development assurance classification at the item level
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
IT	Information Technology; technologies and systems for electronic processing, storage, and transmission of information
MIL-STD	Military Standard; U.S. military standard
MSA	Measurement System Analysis
Nadcap	Industry accreditation program for critical special processes in aviation, space, and defense
OEM	Original Equipment Manufacturer
ONR	ON-Regel; rule published by Austrian Standards
PAM	Process Assessment Model
PDCA	Plan - Do - Check - Act
PL	Performance Level under EN ISO 13849-1
PLD	Programmable Logic Device
PPAP	Production Part Approval Process
PPF	Produktionsprozess- und Produktfreigabe; production process and product approval under VDA
PRI	Performance Review Institute; administrator of the Nadcap program
PRM	Process Reference Model

Abbreviation	Meaning
QM	Quality Management. In ISO 26262, QM denotes a classification outside the ASIL levels.
QMC	Quality Management Center within the German Association of the Automotive Industry (VDA)
QMS	Quality Management System
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals
RoHS	Restriction of Hazardous Substances
RTCA	Radio Technical Commission for Aeronautics
SAE	SAE International
SIL	Safety Integrity Level
SPC	Statistical Process Control
SPICE	Systems/Software Process Improvement and Capability Determination
V&V	Verification and Validation
VDA	Verband der Automobilindustrie; German Association of the Automotive Industry

7 About the Author

Bernhard Wagner, Diplom-Informatiker (Graduated in Computer Science) and MBA, has more than 25 years of experience in software, electronics, systems engineering, and quality management. His areas of focus include functional safety under ISO 26262 and IEC 61508, Automotive SPICE, vehicle diagnostics, and configuration and change management. He has worked as a software developer, systems architect, project manager, and technical director, participated in ASAM and ISO working groups, and co-authored ISO 22900-2, ISO 22900-3, and diagnostics specifications in AUTOSAR.



He also supports companies in assuring the quality of complex electronic systems and implementing standards-compliant development processes.